

LLM Leaderboards

An LLM leaderboard is a public ranking or comparison table that shows how different LLMs perform on a common set of evaluations.

3rd party

Why Leaderboards Exist

- ~~Compare models across labs with a common reference -> also provides trust~~
- ~~Guide model selection when you don't have resources to run evals yourself~~
- Detect saturation (when everyone scores 95%+, it's time for a new benchmark)
- To discover new models

Who needs them, and for what

- AI engineers — shortlisting. Narrow 100 candidates to 5 you'll actually test yourself.
- Frontier labs — competitive positioning and release gating; leaderboard standing is a marketing asset and an internal go/no-go signal.
- Researchers — measuring whether a new technique actually helps, against a shared baseline.
- Policymakers and safety institutes — tracking capability trends for forecasting and regulation.
- Open-weight ecosystem — discovery. A three-person lab with no marketing budget can appear next to a frontier model on the same table. This is genuinely one of the most underrated functions.

Types of LLM Leaderboards

1. Benchmark-Specific Leaderboards

These rank models using the results of one particular benchmark.

For example, a leaderboard may compare models only on:

- MMLU
- HumanEval
- GSM8K
- GPQA

The leaderboard answers:

Which model performs best on this benchmark?

These leaderboards are easy to understand, but they provide only a narrow view of model performance.

2. Multi-Benchmark Leaderboards

These leaderboards compare models across multiple benchmarks and evaluation dimensions instead of relying on one test.

They may measure capabilities such as:

- Knowledge ✓
- Reasoning ✓
- Mathematics ✓
- Coding ✓
- Instruction-following ✓
- Data analysis ✓

→ *lead*

Many multi-benchmark leaderboards also include practical deployment factors such as:

- Cost per token or query
- Response latency
- Output speed
- Context-window size
- Model availability
- Price-to-performance ratio

The results may be displayed as separate scores or combined into an overall ranking.

The leaderboard answers:

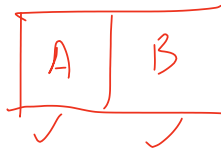
Which model provides the strongest overall combination of capability, cost and performance?

3. Human-Preference-Based Leaderboards

These rank models using human votes.

A user is shown responses from two anonymous models:

- Response from Model A
- Response from Model B



The user selects which response is better.

→ *ranking*

After many such comparisons, models are ranked according to how frequently their responses are preferred.

The leaderboard answers:

Which model's responses do users prefer?

These leaderboards are useful for measuring qualities such as:

- Helpfulness ✓
- Clarity ✓
- Writing quality ✓
- Creativity ✓
- Tone
- Conversational ability

user → better

The limitation is that the preferred answer may not always be the most factually correct answer. People may sometimes prefer an answer because it sounds confident, detailed or polished.

4. Application-Specific Leaderboards

These evaluate models on a particular real-world task or application.

Examples include:

- Fixing software bugs
- Calling APIs and tools
- Generating SQL queries
- Answering medical questions
- Completing agentic tasks
- Working with long documents

coding

The leaderboard answers:

Which model or AI system performs best for this specific application?

Why can't you always trust LLM Leaderboards

1. Benchmark performance may not transfer to real applications

Benchmark questions are usually clean, well-defined and self-contained.

Real applications involve:

- Ambiguous requests ✓
- Missing information ✓
- Company-specific data ✓
- Multiple conversation turns
- Tool failures ✓
- Unusual edge cases ✓

A model that performs well on academic benchmarks may still fail inside a customer-support bot, coding assistant or RAG system.

Leaderboards are therefore useful for shortlisting models, but they cannot replace evaluation on your own use case.

2. Benchmark contamination can inflate scores

Many benchmark questions are publicly available through papers, repositories, tutorials and websites.

These questions may enter a model's training data.

The model may then answer correctly because it has already seen the question, rather than because it can independently solve it.

A high score may therefore reflect:

- Genuine capability ✓
- Memorisation ✓
- Familiarity with similar questions ✓

This is one of the biggest threats to benchmark reliability.

memory

3. Models can be over-optimised for the leaderboard

Once a leaderboard becomes influential, developers begin optimising specifically for it.

They may repeatedly:

1. Submit a model.
2. Study its weak areas.
3. modify the model, prompt or agent.

Eventually, the system may become very good at that leaderboard without becoming equally better on unseen tasks.

This is known as **leaderboard overfitting**.

It reflects a broader principle called Goodhart's Law:

When a measure becomes a target, it becomes less useful as a measure.

4. Overall rankings depend on subjective design choices

multiple
bench

Multi-benchmark leaderboards must decide:

- Which benchmarks to include
- Which capabilities to exclude
- How scores are normalised
- How much weight each benchmark receives
- Whether cost and latency should affect ranking

Changing these choices can change the final ranking.

For example, a coding-heavy leaderboard may favour one model, while a reasoning-heavy leaderboard may favour another.

5. Small score differences may not be meaningful

Leaderboards often show exact ranks even when the difference between models is tiny.

For example:

- Model A: 84.3
- Model B: 84.1

This difference may be caused by:

- Prompt variations
- A small test set
- Evaluation noise
- Automatic-grading errors

A model ranked third may not be meaningfully better than the model ranked fifth.

Rankings become especially unstable when confidence intervals or uncertainty estimates are not shown.

6. Human-preference leaderboards have human biases

Human voting is valuable for measuring helpfulness, tone and writing quality, but people may prefer answers that are:

- Longer
- More confident
- Better formatted
- More agreeable
- More entertaining

The preferred answer may not always be the most accurate, safest or best reasoned.

The voting population may also not represent your users.

Human leaderboards can also be affected by coordinated voting or attempts to identify and favour particular models.

7. The scores may be stale, incomplete or self-reported

A leaderboard may contain:

- Results from older model versions
- Missing newly released models
- Discontinued models
- Self-reported results that were not independently verified

How to Read LLM Leaderboards as an AI Engineer

Leaderboards are a filtering tool, not a decision tool.

1. Write your requirements down before opening any board.

- Task type ✓
- latency budget ✓
- cost ceiling ✓
- context needs ✓
- deployment constraints (on-prem? open weights? data retention?).

Doing this first is what stops you rationalizing toward whatever is ranked #1. If they take one habit away, make it this one.

2. Pick the board that matches your task, not the famous one.

- agents → SWE-bench Pro / Terminal-Bench / OSWorld *embedding*
- Tool use → BFCL
- Chat → the matching LMArena sub-arena.
- RAG → MTEB.
- Budget-constrained → a board carrying price and throughput.

3. Read the board correctly

- What exactly is scored? Answer key, test suite, human vote, or LLM judge?
- Who ran the model? Independent evaluator or vendor self-report?
- At what inference compute? Reasoning-effort settings change both the score and your bill.
- How old is the eval set? Static and public means assume contamination pressure.
- Is it saturated? If the top ten sit within a couple of points, it's over.
- Where are the confidence intervals? No CI means treat adjacent ranks as tied.
- If composite — what are the weights? Unpublished weights mean an unpublished opinion.

4. Cut to 3-5 candidates

Apply the hard filters first: license, price ceiling, latency, availability in your region.

5. Run your own eval. This is where the decision happens.

Fifty to a few hundred examples from your real traffic beats any public benchmark.

Score on quality, cost, latency, and failure behaviour.

SQL query generator

custom

models $\rightarrow$ env